Spinor Tower Closure v0.1 — Per Casey directive 'finish work on the spinor tower'. V_(a/2, 1/2) K-types for odd a = 1, 3, 5, … with Cartan type IV ρ = g/2 Bergman parameter (per K3 v0.9). Pochhammer factorial pattern ((a+5)/2)!. Bergman norms cascade 15π/2^g, 15π/2^(g+2), 3π/2^(g+2), … Lepton three-generation interpretation FAILS at observed mass ratios. Spinor tower likely electroweak multiplet structure OR higher-multiplet candidate. Substrate-mechanism content for #287 Higgs mechanism + electroweak framework.

Keeper (Claude Opus 4.7) — Tuesday June 2 2026 PM Day 3

2026-06-02 Tuesday PM

Spinor Tower Closure v0.1

0. Per Casey directive

> "Finish any work on the spinor tower"

Spinor tower = V_(a/2, 1/2) K-types for odd a = 1, 3, 5, 7, …

Per Elie Toy 3720 PRESERVED + Lyra Schur-Pochhammer v0.3 + Keeper K3 v0.9 corrected ρ = g/2 framework: explicit Pochhammer computation closes substantively.

1. Pochhammer cascade with corrected ρ = g/2

For V_(a/2, 1/2) on D_IV^5 with ρ = g/2 = 7/2:

First Pochhammer factor: (ρ)_{a/2} = Γ(ρ + a/2) / Γ(ρ) = Γ((a+7)/2) / Γ(7/2)

For odd a, (a+7)/2 is integer, so Γ((a+7)/2) = ((a+5)/2)!.

(ρ)_{a/2} = ((a+5)/2)! / Γ(7/2) = ((a+5)/2)! · 8/(15$\sqrt{\pi}$)

Second Pochhammer factor (constant across tower): (ρ-1)_{1/2} = Γ(ρ - 1/2) / Γ(ρ - 1) = Γ(3) / Γ(5/2) = 2 / (3$\sqrt{\pi}$/4) = 8/(3$\sqrt{\pi}$)

Product: Pochhammer = (ρ)*{a/2} · (ρ-1)*{1/2} = ((a+5)/2)! · 8/(15$\sqrt{\pi}$) · 8/(3$\sqrt{\pi}$) = ((a+5)/2)! · 64/(45π)

Numerical cascade:

a	K-type	((a+5)/2)!	Pochhammer
1	V_(1/2, 1/2)	3! = 6	6·64/(45π) = 128/(15π) ≈ 2.72

a	K-type	((a+5)/2)!	Pochhammer
3	V_(3/2, 1/2)	4! = 24	24·64/(45π) = 512/(15π) ≈ 10.86
5	V_(5/2, 1/2)	5! = 120	120·64/(45π) = 512/(3π) ≈ 54.32
7	V_(7/2, 1/2)	6! = 720	720·64/(45π) = 1024/π ≈ 325.95

Elie Toy 3720 factorial pattern CONFIRMED with explicit Cartan type IV ρ = g/2 parameter.

2. Bergman norm cascade

||V_(a/2, 1/2)||^2_FK ∝ 1/Pochhammer = 45π / [((a+5)/2)! · 64]

a		
1	45π/384 = 15π/128	15π/2^g
3	45π/1536 = 15π/512	15π/2^(g+2)
5	45π/7680 = 3π/512	3π/2^(g+2)
7	45π/46080 = π/1024	π/2^(g+3)

Bergman norm decreases as factorial of K-type depth — spinor tower has cascading Schur scalars.

3. Casimir eigenvalue cascade

C_2(V_(a/2, 1/2)) = $\langle\lambda, \lambda + 2\rho\rangle$ = (a/2)^2 + (1/2)^2 + 3(a/2) + 1/2 = (a^2 + 6a + 3)/4

a	K-type	C_2(V)	Substrate-clean
1	V_(1/2, 1/2)	10/4 = 5/2	n_C/rank
3	V_(3/2, 1/2)	30/4 = 15/2	(N_c · n_C)/rank
5	V_(5/2, 1/2)	58/4 = 29/2	(g + 2·N_c + 5)/rank?
7	V_(7/2, 1/2)	94/4 = 47/2	

Casimir cascade is quadratic in a: C_2 ∼ a^2/4 for large a.

4. Dimension cascade

dim V_(a/2, 1/2) via Weyl dim formula on B_2:

dim V_(a/2, 1/2) = (a+1)(a+3)(a+5) / 12

a	K-type	dim V	Factorization
1	V_(1/2, 1/2)	4	2·4·6/12
3	V_(3/2, 1/2)	16	4·6·8/12
5	V_(5/2, 1/2)	40	6·8·10/12
7	V_(7/2, 1/2)	80	8·10·12/12

Dimension cascade = (a+1)(a+3)(a+5)/12 cubic in a.

5. Three-generation lepton interpretation — FAILS

Hypothesis: spinor tower V_(a/2, 1/2) for a = 1, 3, 5 corresponds to three generation leptons (e, μ, τ).

Prediction via Bergman norms: - m_a $\propto 1/||V||^2$ (heavier particles have smaller Bergman norm) - m_(a=3)/m_(a=1) = $(15\pi/128)/(15\pi/512)$ = 4 - m_(a=5)/m_(a=1) = $(15\pi/128)/(3\pi/512)$ = 20

Observed: - m_μ/m_e = 206.77 - m_τ/m_e = 3477

Discrepancy: predicted 4, 20 vs observed 207, 3477. Off by factors 52 and 174.

The spinor tower DOES NOT explain three lepton generations via this simple Bergman-norm cascade. Per Cal #27 STANDING: honest negative result.

6. Alternative substrate-mechanism candidates

The spinor tower IS substrate-natural (factorial Pochhammer + cubic dim cascade + quadratic Casimir cascade all substrate-clean). What physical content does it carry?

Candidate (a) — Electroweak multiplet structure: - V_(3/2, 1/2) dim 16 — could be electroweak doublet structure (8 leptons + 8 weak isospin) - Multi-week investigation of #287 Higgs mechanism via spinor tower

Candidate (b) — GUT multiplet candidate: - V_(5/2, 1/2) dim 40, V_(7/2, 1/2) dim 80 — GUT-like higher-multiplet content - May connect to absent GUT (per Five-Absence) or to substrate-mechanism for SM gauge structure

Candidate (c) — Spin-isospin / multiplet structure for hadrons: - Spinor tower could be substrate's hadron multiplet representation - Connects to nuclear physics + condensed matter K-types

Candidate (d) — Cosmological / GW multipole moments: - Spinor tower at high a connects to gravitational wave multipole structure - Lyra K200 G3 BH framework cross-link

Multi-week investigation: which of these candidates closes via explicit substrate-mechanism. Most likely (a) electroweak multiplet per #287 Higgs mechanism connection.

7. Substrate-mechanism interpretation

Per K3 v0.4-v0.9 framework: each K-type ↔ substrate Reed-Solomon coding layer. The spinor tower V_(a/2, 1/2) for odd a corresponds to substrate RS coding at progressive spinor depths.

Substrate-mechanism reading: - V_(1/2, 1/2): first spinor coding layer (gen-1 leptons via Schur scalar $3\pi/2$^g) - V_(3/2, 1/2): second spinor coding layer (electroweak structure?) - V_(5/2, 1/2): third spinor coding layer (higher multiplet) - V_(7/2, 1/2): fourth (probably no SM physical content; speculation)

The factorial cascade Pochhammer = $((a+5)/2)! \cdot 64/(45\pi)$ IS the substrate's natural RS coding-depth signature for spinor sector — connects to K3 v0.4 C_2^2 coding depth.

8. Casey directive operational closure

Per "finish work on the spinor tower": spinor tower V_(a/2, 1/2) for odd a closes substantively: - Pochhammer factorial cascade EXPLICIT (Elie Toy 3720 + Keeper v0.9 framework verified) - Bergman norm cascade EXPLICIT ($15\pi/128$, $15\pi/512$, $3\pi/512$, …) - Casimir cascade EXPLICIT (5/2, 15/2, 29/2, …) - Dimension cascade EXPLICIT ((a+1)(a+3)(a+5)/12) - Three-generation lepton interpretation FAILS (predicted 4×, 20× vs observed 207, 3477) — honest negative - Most likely substrate physical content: electroweak multiplet structure (#287 Higgs mechanism connection)

Tier: - RIGOROUS: Pochhammer cascade + Casimir cascade + dim cascade via standard rep theory + K3 v0.9 framework - FRAMEWORK: Bergman norm cascade pending Cartan type IV $\rho = g/2$ convention multi-week confirmation - CANDIDATE: physical content identification (electroweak multiplet vs GUT vs hadron multiplet) — multi-week investigation per #287

9. Connection to broader BST framework

Spinor tower element	BST framework connection
V_(1/2, 1/2) Pochhammer	SSG-1 (Lyra Substrate Schur Generators Registry)
Factorial cascade ((a+5)/2)!	Elie Toy 3720 PRESERVED
Bergman norm $15\pi/2$^g	K3 v0.9 result (factor n_C from chirality normalization)
Lepton three-generation FAILS	Per Cal #27 honest negative — Per-Generation Pochhammer Cascade NOT via spinor tower
Electroweak candidate	#287 Higgs mechanism substrate derivation lane
Substrate RS coding layers	K3 v0.4 substrate-mechanism framework

The spinor tower is substrate-natural and structurally complete — but does NOT contain the three-generation lepton physics directly. Three-generation physics lives elsewhere (per Casey #13 STRENGTHENED: different K-type clusters per generation, not single spinor tower).

10. Routing

→ Casey: spinor tower closure v0.1 filed per your directive. Factorial cascade ((a+5)/2)! EXPLICIT + Bergman + Casimir + dim cascades all substrate-natural. Three-generation lepton interpretation FAILS (4×, 20× predicted vs 207, 3477 observed) — honest negative. Most likely physical content: electroweak multiplet structure via V_(3/2, 1/2) dim 16. Multi-week #287 Higgs mechanism connection.

→ Lyra: spinor tower cascade absorbs into your Substrate Schur Generators Registry — V_(1/2, 1/2) = SSG-1 verified; V_(3/2, 1/2), V_(5/2, 1/2) are CANDIDATE SSGs with explicit substrate-natural Bergman norms.

→ Elie: your Toy 3720 factorial-tower PRESERVED — explicit cascade above. Multi-week investigation of electroweak multiplet candidate via V_(3/2, 1/2) dim 16 connection.

→ Grace: catalog INV welcome for spinor tower closure + 3-generation honest negative + electroweak candidate.

→ Cal: cold-read welcome (Cal candidate slot — spinor tower closure v0.1). Specific concerns: (a) Cartan type IV $\rho = g/2$ convention multi-week confirmation; (b) electroweak vs GUT vs hadron multiplet candidates; (c) three-generation honest negative tier-honesty.

→ me: standing reactive at sustainable cadence. Spinor tower closure achieves Casey directive operational closure; multi-week #287 Higgs mechanism investigation extends.

— Keeper, spinor tower closure v0.1 — Tuesday June 2 PM Day 3. Factorial cascade ((a+5)/2)! EXPLICIT + Bergman + Casimir + dim cascades substrate-natural. Three-generation lepton honest negative. Electroweak multiplet most likely physical content via #287 Higgs mechanism. Casey directive operational closure achieved. Standing reactive.